SE444: AI Mid-Term II - Compact Grading Solutions — Total: 20 points — Time: 90 min — Prof. Anis Koubaa

Part I: Multiple Choice (4 pts)

Q	A	B	C	D	E
I.1				X	
I.2			X		
I.3	X	X	X		X
I.4		X		X	

Q1 (FOL - 1pt): Given $\forall x(P(x) \rightarrow Q(x))$, $\exists x P(x)$. Which entailed?
Opts: A) $\exists x Q(x)$ B) $\forall x Q(x)$ C) $\exists x(P(x) \wedge Q(x))$ D)$\forall x(Q(x) \rightarrow P(x))$ E) $\exists x(Q(x) \wedge \neg P(x))$
Ans: A, C (0.5pt each)

Q2 (Prop Logic - 1pt): KB: $(p \wedge q) \rightarrow r$, $r \rightarrow (s \vee t)$, $\neg s \wedge \neg t$, q, $p \vee \neg r$. Which true?
Opts: A) Unsatisfiable B) $\models p$ C) $\models \neg p$ D) $\models (r \wedge s)$ E) Satisfiable with $p = T, r = T$
Ans: C (1pt)

Q3 (CSP AC-3 - 1pt): Vars A, B, C. Domains: $D(A) = \{1, 2\}$, $D(B) = \{1\}$, $D(C) = \{1, 2\}$. Constraints: $A \neq B$, $B < C$. After AC-3?
Opts: A) $D(C) = \{2\}$ B) $D(A) = \{2\}$ C) All singletons D) $B \leq C$ also singletons E) $D(B) = \{1, 2\}$ still prunes C to $\{2\}$
Ans: A, B, C, E (0.25pt each)

Q4 (Agents - 1pt): Delivery drones (partial, stochastic, dynamic). Which correct?
Opts: A) Simple reflex w/ large table rational B) Model-based better than simple reflex C) Goal-based needs determinism D) Utility-based values info gathering E) Table-driven always optimal
Ans: B, D (0.5pt each)

Part II: Logical Reasoning (10 pts)

Predicates: Student(x), Teacher(x), Course(x), Takes(x,y), Teaches(x,y), Understands(x,y), Topic(y), Covers(c,y), GradedBy(x,t), Passes(x,y)

Q1 (FOL Translation - 3pts): "Every student who takes a course taught by a teacher understands at least one topic that the course covers."
Ans: $\forall s, t, c\,[\text{Student}(s) \wedge \text{Teacher}(t) \wedge \text{Course}(c) \wedge \text{Takes}(s, c) \wedge \text{Teaches}(t, c) \rightarrow \exists u\,[\text{Topic}(u) \wedge \text{Covers}(c, u) \wedge \text{Understands}(s, u)]]$
Grading: 3pts for correct structure (quantifiers, implication, existential)

Q2 (Negation Translation - 3pts): "No teacher grades a student who is not enrolled in one of their courses."
Ans: $\forall t, s\,[\text{Teacher}(t) \wedge \text{Student}(s) \wedge \text{GradedBy}(s, t) \rightarrow \exists c\,[\text{Course}(c) \wedge \text{Teaches}(t, c) \wedge \text{Takes}(s, c)]]$
Grading: 3pts for correct implication form

Q3 (CNF - 4pts): Negate & convert to CNF: $\forall x\,[\text{Student}(x) \rightarrow \text{Hardworking}(x)]$
Steps & Ans: 1. $\neg\forall x\,[\text{Student}(x) \rightarrow \text{Hardworking}(x)]$ 2. $\exists x\,\neg[\text{Student}(x) \rightarrow \text{Hardworking}(x)]$ 3. $\exists x\,[\text{Student}(x) \wedge \neg\text{Hardworking}(x)]$ 4. Skolemize: $\text{Student}(a) \wedge \neg\text{Hardworking}(a)$ (CNF)
Grading: 1pt/step, 1pt for final CNF

Part III: CSP (6 pts)

Context: 3 labs, 3 TAs. Overlaps: Lab1(8-9) & Lab2(8:30-9:30), Lab2 & Lab3(9-10)

Q1 (Formulate - 2pts): Variables, domains, constraints?
Ans: • Vars: L_1, L_2, L_3 (0.5pt) • Domains: $D(L_1) = \{A, C\}$, $D(L_2) = \{A, B, C\}$, $D(L_3) = \{B, C\}$ (1pt) • Constraints: $L_1 \neq L_2$, $L_2 \neq L_3$ (0.5pt)

Q2 (AC-3 - 3pts): Apply arc consistency. Final domains?
Ans: No pruning. All domains unchanged: $\{A, C\}, \{A, B, C\}, \{B, C\}$ (1.5pts)
Process: Check arcs (L_2, L_1), (L_1, L_2), (L_2, L_3), (L_3, L_2) - all values supported.
Grading: 1.5pts no pruning, 1pt work shown, 0.5pt final domains

Q3 (Solution - 1pt): Valid assignment?
Ans: e.g., $L_1 = A, L_2 = B, L_3 = C$ or $L_1 = A, L_2 = C, L_3 = B$ or $L_1 = C, L_2 = A, L_3 = B$ or $L_1 = C, L_2 = B, L_3 = C$ (1pt)

Grading Notes: Partial credit for logical reasoning & work shown. Total: 20 points